

# Fatemeh Shahrokhshahi

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## RESEARCH INTERESTS

Developing computational methods to improve reasoning and reliability in large language models, with a focus on failure prediction, structured inference, and prompt-based intervention strategies.

## EDUCATION

**Master of Computer Engineering** *Istanbul Aydin University, Istanbul, Turkey* | GPA: 4.0/4.0 Sep 2024 - Present

**Bachelor of Robotic Engineering** *Shahrood University of Technology, Shahrood, Iran* | GPA: 2.62/4.0 Sep 2015 - Feb 2022

## PUBLICATIONS

**LogiCue: Targeted Prompting for Improved Modal and Conditional Reasoning in LLMs** | Dec 2025

*Procedia Computer Science, Elsevier* | *ACLing 2025* | *Published*

- Pattern-specific prompting framework for seven modal inference patterns, evaluated across nine models with statistical validation ( $p < 0.01$ )

**Can We Predict LLM Reasoning Failures? Structural Predictability of Modal and Conditional Inference Errors** | Aug 2026

*KI 2026: 49th German Conference on Artificial Intelligence* | *Published, Bremen, Germany*

- Trained classifiers to predict per-instance LLM failures from input-side structural features (3,776 instances, nine models, three prompting strategies; AUC-ROC 0.69–0.93); showed targeted prompting and output-side reliability mechanisms are complementary

**Towards Logical Reasoning by Design: Encoding Inference Rules in MeMo's Associative Memory** | Jun 2026

*MeMo Workshop on Mechanistic Interpretability & Neuro-symbolic Approaches, University of Rome Tor Vergata* | *Accepted*

- Explored explicit rule storage in MeMo's Correlation Matrix Memory.

## RESEARCH EXPERIENCE

**Intelligent Academic Retrieval System (RAG)** Nov 2024 – Feb 2025

RAG pipeline over 512 Springer papers; embedding-based retrieval outperformed keyword search [GitHub]

**Balanced K-Means for Domain Discovery in LMs** Feb – May 2025

Balanced k-means clustering preventing general-domain dominance on C4 dataset

**Earthquake Prediction Using ML (Under Prep.)** Jul 2025

Ensemble of Random Forest, PINNs, GNNs on AFAD data; 76.9% accuracy / 82.0% AUC; 90-day horizon

**Coefficient Sensitivity and Domain Robustness of Cross-Lingual Steering (Pilot)** Sep 2026

DiffMean-style activation steering (Llama-3.2-1B) testing whether steering robustness under domain shift depends on coefficient calibration rather than input domain; found a narrow, sharply bounded effective coefficient window consistent across domains [GitHub]

**Child Mental State Analysis via Drawing Behaviour** 2019 – 2021

Android app capturing stroke/symbol data; output structured features to a clinical analysis system

## TECHNICAL SKILLS

**Programming:** Python, C, Java, MATLAB, SQL, FastAPI

**Deep Learning Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain

**Machine Learning:** Representation Learning, Deep Learning, Bayesian Methods

**NLP/AI Specializations:** Large Language Models, Natural Language Processing, Knowledge Graph Embeddings

**Research Tools:** Git, Cloud Services (AWS/GCP), ChromaDB (Vector DB), Julep (Orchestration), Model Fine-tuning

**Additional:** Computer Vision (OpenCV), Statistical Analysis, Experimental Design

## CERTIFICATIONS & SERVICE

**IELTS Academic** | *TRF 22IR007446SHAF010A* Oct 2022

- Overall: 7.0 (L:8, R:7, S:7, W:6) | Persian (native) · English (C1) · French (A2) · Turkish (basic)

**Academic Service** |

- Reviewer, NeurIPS 2026 Workshop on Verify-Agents Sep 2026
- PC Member, EITS 2026 Apr 2026
- Presenter, ACLing 2025 (British University in Dubai) Dec 2025